Fiscal Year:	FY 2006	**Task Last Updated:**	FY 05/04/2006
PI Name:	Goree, John Ph.D.		
Project Title:	Optically-Excited Waves in 3D Dusty Plasmas		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Complex fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	john-goree@uiowa.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	319-335-1843
Organization Name:	University of Iowa		
PI Address 1:	Department of Physics and Astronomy		
PI Address 2:	210 Van Allen Hall		
PI Web Page:			
City:	Iowa City	**State:**	IA
Zip Code:	52242-1403	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Fluid Physics 01-OBPR-02: Research and Flight Experiment Opportunities
Start Date:	04/01/2002	**End Date:**	03/31/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:	Rogers, Richard	**Contact Phone:**	216-433-6512
Contact Email:	Richard.B.Rogers@grc.nasa.gov		
Flight Program:			
Flight Assignment:	NCE until 3/31/2007 per PI (5/2006)		
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):			
Grant/Contract No.:	NAG3-2771		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Original Abstract: Experiments will be carried out in an interdisciplinary study of waves (i.e., phonons) in 3D Coulomb lattices. The experimental system is a suspension of charged polymer microspheres in a gas-discharge plasma. This so-called "dusty plasma" is a macroscopic system where the microspheres arrange themselves, due to their mutual repulsion, in a lattice. Shear waves will be excited in the lattice using the radiation pressure of a laser beam. New Strategic Abstract: During the final two years of this grant, the research goals are shifted to emphasize strategic goals related to NASA's planned manned space flight missions to the Moon and Mars. Countermeasures for dust contamination in Advanced EVA will be the theme. Experiments will be performed to test materials and investigate processes related to dust adhesion on surfaces and the use of plasma exposure to remove dust from those surfaces. Original Objective: We will prepare for future flight experiments, which will have two tasks: * We will study the dispersion relation (i.e., phonon spectrum) of sinusoidal waves and the propagation of pulsed waves. These experiments will be performed with plane waves. * We will study the formation of Mach cones, which are the V-shaped wakes or shocks created by a supersonic disturbance. In this experiment, a moving laser spot is the supersonic disturbance. New Strategic Objective: Preliminary experiments will be carried out to investigate adhesion properties of JSC-1 Lunar Simulant dust to EVA suits, and the effectiveness of plasmas in removing that dust. Original description: * Need for microgravity: These experiments are an extension of 2D experiments performed in the laboratory, under gravity conditions. Sedimentation due to gravity causes the microspheres to settle rapidly into a 2D layer that is levitated by the strong electric field near an electrode in a plasma. By eliminating gravity, it is possible to fill a 3D volume with the suspension. * Use of a novel experimental system: Our experiments will exploit the low damping of a dusty plasma as compared to colloidal suspensions. We will excite waves using a manipulation laser, which applies a radiation pressure on the particles. Using a dusty plasma allows us to measure particle motion in-situ using direct imaging with video microscopy. * Our scientific effort and our team are interdisciplinary. New Strategic Task Significance: Advanced EVA has been identified as one of the four critical technologies for NASA's New Vision for Space Exploration. The EVA System includes EVA suits and airlocks, for which dust contamination is a risk. According to Lara Kearney, JSC's manager of Advanced EVA, 13 challenges of present-day technologies have been identified, and two of these include dust: the present-day EVA suit protection from dust intrusion is inadequate, and dust on airlocks is an unsolved risk. Our project will include studies for low-TRL technology needs related to dust on the Moon and Mars. “Iowa Dust Mitigation Scheme”: We performed experiments at The University of Iowa to test methods of cleaning space suits. In our “Iowa Dust Mitigation Scheme,” the crew of a lunar habitat would use two steps to clean their suits after returning to the habitat from an EVA. Firstly, they will use air to blow dust off the suits during airlock pressurization, and secondly, after egress from the suit, they will use plasma to electrostatically clean the residual dust. These two steps have the following advantages: (1) they can both be done in the airlock, with dust collecting at the bottom of the airlock, thereby isolating it from entering the habitat; (2) they will not require any water, which will be a precious resource; (3) they do not involve any frictional rubbing of the suit’s fabric, which would reduce the suit’s lifetime. 1st step of the “Iowa Dust Mitigation Scheme”: We exploit the availability of air flow during pressurization of the airlock. We propose that air should not be introduced into the airlock diffusely, but instead it should be introduced through a hose that the astronaut will direct onto the spacesuit, while still wearing the suit. This will blow off a large fraction of the dust, which would collect at the bottom of the airlock where it can be emptied periodically. 2nd step of the “Iowa Dust Mitigation Scheme”: The second step will exploit the availability of vacuum conditions in an airlock. After the astronaut has removed the space suit and egressed, the suit will hang in the airlock. The airlock will then be partially evacuated, with a gas pressure of approximately 1 mBar. Radio-frequency electrical power will be applied, ionizing the gas to make plasma that will be exposed to the airlock’s contents, including the suit. Electrostatic effects will remove dust from the suit. This method was first introduced by T.E. Sheridan et al., J. Geophys. Res. 97, 2935 (1992), and it has been used in the semiconductor industry for cleaning particulates from silicon wafers. While our scheme is intended to be used in a lunar Habitat, it could also be adapted to a habitat on Mars, which will also require airlocks. The electrostatic effects that will remove dust from the suit will be the primary focus of this research. When a particle rests on a surface, it is held in place by a van-der-Waals attraction, but this adhesion can be overcome by several mechanisms. One such mechanism for the Moon and Mars environments is the application of electrostatic forces. Scientific results from this project will also find other applications: (1) One issue is cleaning solar panels on Mars. Dust storms on Mars will gradually cover the panels, reducing their efficiency, unless they are cleaned. Because Mars has a very low-pressure atmosphere, it may be possible to use radio-frequency power to ionize the Martian atmosphere immediately above a solar panel, and thereby clean it. (2) Another issue is gaining an understanding of dust particle leviation and transport on the Moon. The Moon, unlike Mars, has no atmosphere. Instead, it is immersed in interplanetary plasma, which consists of free electrons and protons at a low number density. The Moon disturbs this plasma, so that there is an electric field perpendicular to the Moon’s surface. This electric field applies a vertical force to any electrically-charged fine particle. Particles are known to become charged and levitated on the Moon. This was discovered in Apollo missions, where a thin cloud of dust was seen floating above the lunar horizon. Particles can be charged either photoelectrically by exposure to solar UV, or by collection of free electrons and ions from the surrounding plasma. This can lead to levitation, if the particle can first overcome van-der-Waals adhesion to surfaces such as a rock, a photovoltaic cell, or another dust particle.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	

Task Progress:

During the current reporting period, April 2005 – March 15 2006:
1. Detailed results were reported for our first series of experiments.

This first series of experiments included tests of both our plasma method and our airflow method for dust mitigation. In both methods, we used actual spacesuit fabric that we coated with JSC-1 lunar simulant, and we used video cameras to observe dust flying off the spacesuit fabric during the cleaning process. For each experimental run, we used a freshly prepared sample which was covered with dust by shaking the sample in a bag with the dust.

The plasma method was carried out using a vacuum chamber, operated at a reduced gas pressure, as could be achieved either in an airlock or in the Martian atmosphere. Treatment conditions were a gas pressure of 0.1 mTorr and room temperature. The gas pressure was steady, and a high voltage was suddenly applied to an electron source to start a plasma, which was thereafter held under steady conditions. This plasma filled the entire vacuum chamber, so that the sample, which was inserted into the middle of the chamber, was exposed to the plasma.

The plasma treatment method was shown to cause dust to fly off the spacesuit fabric. In this preliminary test, measurements were not quantitative, and were presented visually as videos and still images from the video. (Quantitative results are presented in the second series of experiments described later.)

The airflow cleaning method was tested in a different vacuum chamber, a bell jar. This bell jar was evacuated, and then air was admitted to the chamber through a nozzle that was pointed at the sample surface. The same sample was used as in the plasma treatment. The bell jar had a volume of 1.3 cubic feet, as compared to 150 cubic feet for the Space Shuttle airlock. The nozzle had an inside diameter of 0.157". The airflow began when we suddenly opened a valve between the nozzle and a tube leading to the atmosphere. Air flowed through the nozzle, filling the chamber to atmospheric pressure in 35 seconds, so that the average flow rate was 65 meters/sec. The average airflow velocity during this treatment was 87 meter/sec, which is a velocity that can realistically be attained in an airlock.

We found that the airflow method was capable of cleaning the surface rapidly. Most of the cleaning occurred during the first second of applying airflow. The most significant limitation that we observed was that dust recirculated in the chamber and deposited again on the sample surface during the cleaning process; we expect that this limitation could be overcome by simply including an airfilter in the chamber. As tested, the chamber had only smooth glass and metal surfaces, with no baffles or filtering elements.

We presented these results for the first series of experiments in two talks, both entitled “Iowa Dust Mitigation Scheme for EVA Suits in a Lunar Habitat” . First, the work was presented at the NASA Dust Mitigation Technology Focus Group Workshop, in Golden, CO in June 2005. At this workshop, the PI also participated in focus groups that wrote recommendations for future NASA projects related to dust mitigation. Second, the work was presented at the International Conference on Dusty Plasmas, held in Orleans, France in June 2005. The poster for these talks is also widely disseminated on a web page on the PI's web server: http://dusty.physics.uiowa.edu/

2. We carried out a second series of experiments to further develop the plasma cleaning method.

This work was done by graduate student Timothy Flanagan, working with the PI. It will be reported in a paper that is now being prepared for publication.

The sample preparation method used here was different from the first series. In this series, we again used JSC-1 lunar simulant, but instead of spacesuit fabric here we used a glass surface. This glass surface, which was a sphere, was uniformly coated with JSC-1 simulant using an electrostatic method under atmospheric conditions. We then waited 24 hours, so that any residual electric charge on the sample was dissipated in the atmosphere, and we then had a dusty glass surface with no electric charge. We inserted this sample into our vacuum chamber, and then evacuated the chamber using pumps. The sample was then ready for the test.

In this second series of experiments, we tested three configurations for plasma treatment. In one configuration, there was both a plasma and an electron beam. This beam had a low energy of 70 electron volts. In the other two configurations, we operated with a plasma but no beam, and a beam with no plasma.

Four diagnostics were used in this test. First, as in the first series of expeirments, we recorded a video of the particles flying off the surface. This was done by using a video camera viewing the space immediately above the sample surface. We illuminated the empty volume above the sample surface using a powerful 1 Watt argon laser beam. Second, we recorded a time series of data for the cleaning rate by imaging the same empty volume above the sample using a photomultiplier tube (PMT). When a dust particle flew through the laser beam above the sample, laser light was scattered diffusely, and this was detected by the PMT. We recorded a time series of the laser-light scattering (LLS) signal. Third, we weighed the dust applied to the sample, so that we knew the quantity of dust that was applied. Fourth, we photographed the sample before and after treatment so that any cleaning effect could be easily recognized.

We found that the plasma-plus-beam configuration results in an effective and rapid cleaning of the surface. The plasma-only and beam-only configurations, on the other hand, did not have a cleaning effect, indicating that it is necessary to operate with both the plasma and beam operating simultaneously.

By increasing the plasma density, it is possible to almost completely clean lunar simulant dust from a glass surface within two seconds.

From our LLS time series, we found that the rate of particle release from a surface is related to the square of the plasma density. This finding will be helpful for NASA in predicting the required operating parameters for a plasma source, for cleaning surfaces at lunar and Martian habitats.

3. Plans to continue this work:

The report above describes the work done during the 12-month period ending March 31, 2006.

The project will continue in a no-cost extension until Mar. 31, 2007. During this extension, we plan to carry out the following tasks. We will write a paper for submission to a refereed journal describing the plasma cleaning method. Mr. Flanagan will complete his master's thesis. We will send a copy of our results to the Advanced EVA Systems office at NASA-JSC, and we will disseminate our results on our webpage.

After the completion of this project, the PI will be available for further work on dust mitigation, if any NASA funding materializes.

Bibliography Type:	Description: (Last Updated: 01/23/2025)
Abstracts for Journals and Proceedings	Goree J. "Iowa Dust Mitigation Scheme for EVA Suits in a Lunar Habitat." Presented at the NASA Dust Mitigation Technology Focus Group Workshop, Golden, CO, June 2005. NASA Dust Mitigation Technology Focus Group Workshop, June 2005. http://dusty.physics.uiowa.edu/%7Egoree/talks/NASA%20EVA%20workshop%20Golden%20CO%20May%202005%205.pdf , Jun-2005